

Source boards based on Arduino, and teach students robotics through it. In a few years' time, Arduino will become a standard platform for teaching micro-controllers and robotics to students in colleges."

Another popularly-used platform in India is the Beagle Board—an interesting piece of open hardware developed by a skilled community including several employees of Texas Instruments (TI). The Beagle Board is a low-cost, fanless, single-board computer based on low-power TI processors featuring the ARM Cortex-A8. It is known for being light but highly expandable. Jayakrishnan informs that Energid uses the Beagle Board for remote-control video processing in some of its products.

BITS-Pilani's Centre for Robotics and Intelligent Systems (CRIS) has developed microcontroller hardware for use by students in their courses.

"All designs, software and source code are released under the Creative Commons Attribution Share-Alike (India) licence for use by students of BITS and anyone else," says de Sa.

A lot of Open Source software is also used in robotics. We have already mentioned some software platforms such as ROS, OpenCV and Urbi. OpenRAVE is another Open Source, cross-platform, plug-in-based robot planning architecture that serves all planning and execution purposes. Developed at Carnegie Mellon University, it includes services like collision detection, robot kinematics, physics, robot controls and a network scripting environment.

OpenJAUS and RI-JAUS software development kits used in the joint

An Open Source model for your business

Being a nascent industry, it would be worthwhile for robotics start-ups to consider an Open Source model for their business. This is also quite profitable as can be seen from the success stories of Red Hat and Canonical (Ubuntu) in the software field.

There are tried and tested business models, specifically for open hardware. These include the Forty Percent Model, the Third-Party Catch and Semi-transparent Pricing model, etc. Read more about these at www.longtail.com/the_long_tail/2009/01/a-business-mode.html

architecture for unmanned systems include several useful code components in C++, as well as software to standardise an unmanned system automatically. Orocos is an Open Source tool chain for real-time robotics that covers all aspects of robot control. Orca is an Open Source framework for developing component-based robotic systems. Player is another tool to create software architectures for robotics and sensor systems. It also provides simulation platforms Stage and Gazebo for testing the tools.

The Rossum Project (<http://rosum.sourceforge.net/>)—an attempt to collect, develop and distribute software for robotics applications—is a great starting point for those looking for Open Source tools for their robotics projects.

It is just the beginning

Summing up the current situation, Jayakrishnan says, "While some specialised tools and platforms are available, currently most of the available Open Source robotics components are being borrowed from other fairly mature fields such as radio-controlled modelling, multimedia, computer gaming and professional toys. Entries to robotic games such as Micromouse and RoboSoccer currently use a lot of Open Source components. Another trend is to build a generic platform targeting a particular area, say, wheel robots, and then make the software Open Source. Surveyor Corporation's SRV-1 robot is an example."

Open Source in robotics is sure to strengthen in the future enabling much advancement in robotics. This requires the community to give back as much as it consumes. This can be done by starting new Open Source projects

and uploading your designs, or by improving the hardware and software belonging to your existing projects. You can contribute CAD files for hardware components, Gerber files in the case of PCB layouts, Initial Graphics Exchange Specification (IGES) files for mechanical components, and software code—for new and existing Open Source projects.

Azad says, "Right now, the Open Source community in robotics is at a very nascent stage and a lot of organisations are coming up with their own standards. So some components and codes are incompatible. There will be in time one unified platform like Ubuntu or Fedora, which will be universally accepted by all. Once the software is set, the hardware will be developed accordingly as per the performance of the robot and the types of sensors needed to be integrated. This will help develop a plug-and-play scenario for sensors, actuators, hardware and software."

In all, Open Source seems to hold a lot of promise for the future. However, de Sa feels, "It will take a while to catch on in India where people are still wary of releasing their designs to the public. It takes a certain level of mutual trust to say 'I'll share with you; please share with me.'"

Jayakrishnan concludes by saying that both Open Source and closed source have roles to play in robotics. "I think it is important to realise that it isn't one over the other, but how both together contribute to the advancement of robotics. Open Source is more accessible and lower-cost, but large, centrally-organised investments can be made in closed source, allowing faster development," he says. ●

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Wish-list

While the Open Source repertoire has a lot of tools for the robotics community, there is more to be wished for:

- Custom mechanical hardware
- Sensor technology (especially light detection and ranging sensors)
- Modular robot actuators
- Common input power, common communication and control protocols
- Modular, reconfigurable robot designs